7 Proportions and percentage change

Sheet 39 Tue 2 Feb. 2027 *Proportions and percentages; proportion of a proportion*

1. Warm-up · Quick Calculate 15 % of 240.
2. Core · Method 30 % of the pupils are boarders and 40 % of the boarders do dance. What proportion of the pupils are boarders who do dance?
3. Core + · Problem A school has 800 pupils, of whom 55 % are girls; 20 % of the girls study Latin. How many girls study Latin?
4. Review · Quick 3 exercise books cost 4.50 €. Price of 7 books?
5. Thread · True/false True or false? Justify. The number -2 is a solution of the equation $3x + 1 = -5$.

Sheet 40 Thu 4 Feb. 2027 *Absolute change, multiplier, rate of change*

1. Warm-up · Quick A price goes from 80 Ar to 92 Ar. What is the absolute change?
2. Core · Method For the same change, give the rate of change and the multiplier.
3. Core + · MCQ A decrease of 20 % corresponds to the multiplier:

   A. 0.2 **B.** -0.8 **C.** 1.2 **D.** 0.8
4. Review · Quick Temperatures -2, 0, 3, 5, 9 (in °C). Mean and range?
5. Thread · Quick Calculate the volume and the total surface area of a cube with edge 3 cm.

Sheet 41 Mon 8 Feb. 2027 *Successive percentage changes*

1. Warm-up · Quick Calculate 1.2×0.9.
2. Core · Method An increase of 20 % is followed by a decrease of 10 %. What is the overall rate of change?
3. Core + · True/false True or false? Justify. An increase of 50 % followed by a decrease of 50 % returns to the initial value.
4. Review · Quick Calculate $x^2 - x$ for $x = \dfrac{1}{2}$.
5. Thread · Write-up $ABCD$ is a parallelogram such that $AC = BD$. What type of quadrilateral is it? Justify in one sentence.

Sheet 42 Tue 9 Feb. 2027 *Reverse percentage change; proportion and change*

1. Warm-up · Quick Calculate $\dfrac{1}{1.25}$.
2. Core · Method After an increase of 25 %, what decrease brings back the initial value?
3. Core + · Interpret Out of 250 candidates, 150 are successful; last year there were 200 candidates. Calculate the proportion of successful candidates, then the rate of change of the number of candidates.
4. Review · Table Table of values of f:

x	0	1	2	3
$f(x)$	4	1	0	1

   Image of 2? Preimages of 1?
5. Thread · MCQ On the globe, two points on the same parallel have:

   A. the same latitude **B.** the same longitude **C.** the same distance from Greenwich **D.** the same altitude